

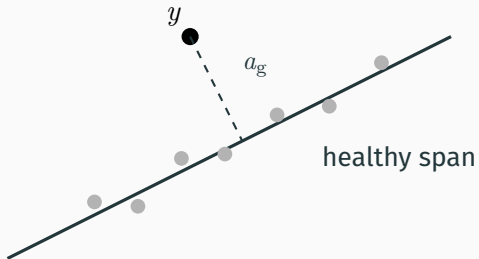
Global and Local Detectors as Two Beliefs

One residual, two assumptions about healthy data

Hankel_AD

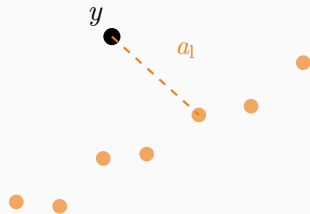
2026-10-07

Two detectors



global

distance to the span

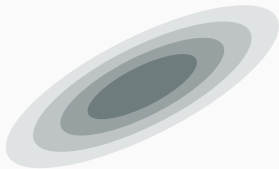


local

distance to nearest training window

One quantity, two beliefs

$$a(y) = \left\| y - \mathbb{E}_\mu[X \mid y] \right\|$$



$$\mu = \mathcal{N}(0, C)$$

global

VS

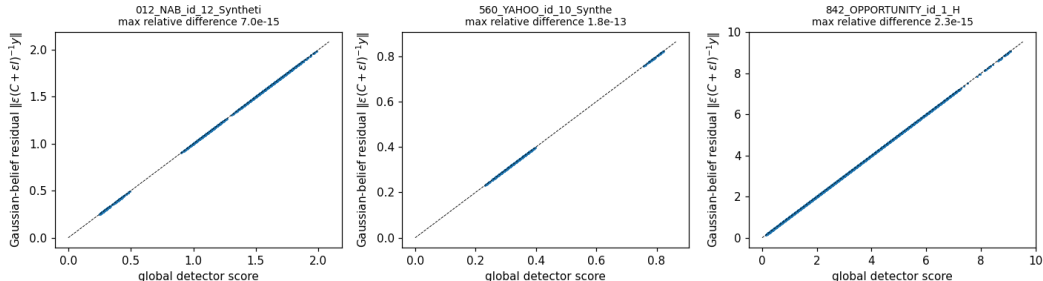


$$\mu = \frac{1}{n} \sum_i \delta_{x_i}$$

local

Check 1: global = Gaussian-belief residual

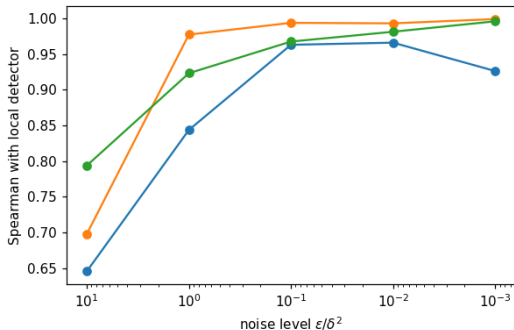
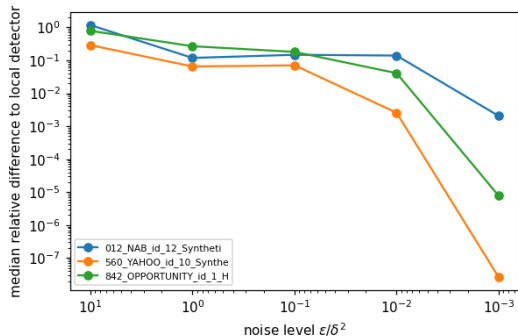
Proposition 1: the global detector equals the Bayes denoising residual under a single Gaussian belief



largest relative difference 1.8×10^{-13}

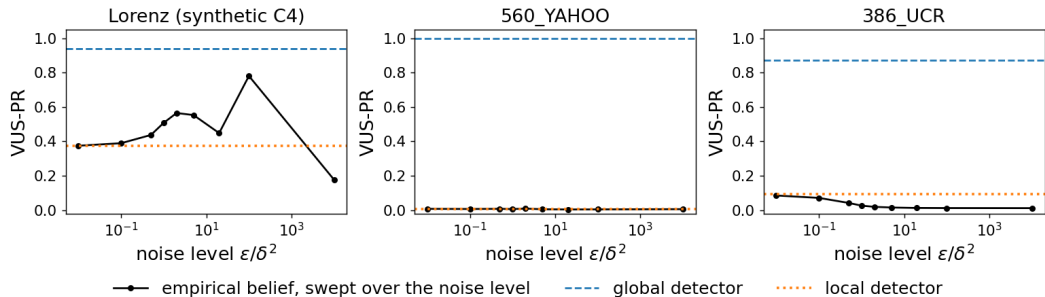
Check 2: local = limit of the empirical-belief residual

Proposition 2: the empirical-belief residual tends to the nearest-neighbour distance as the noise level shrinks (test windows)



converges as noise $\rightarrow 0$ $10^{-8} - 10^{-3}$

The belief, not the noise level

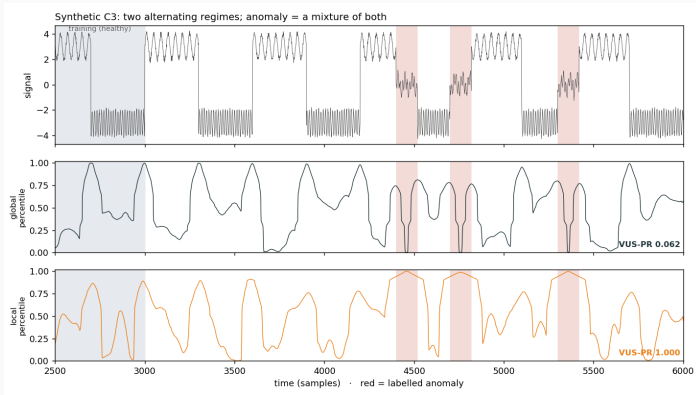


never reaches **global**

What each belief implies

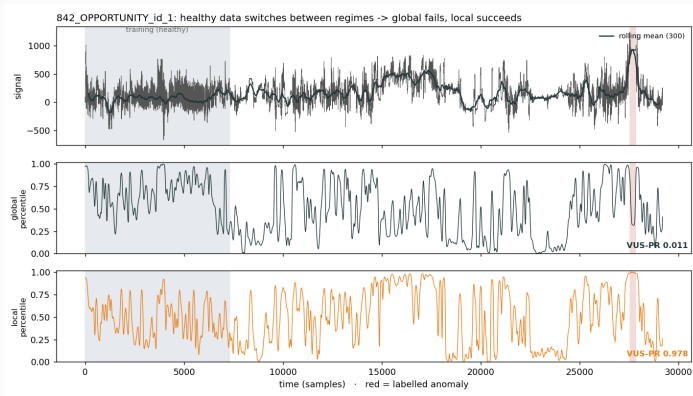
	global	local
outside the span	sees	sees
in span, off training set	blind	sees
healthy but unseen	tolerates	false alarm

In span, off training set: a mixture of regimes



global **0.08** local **1.00** VUS-PR, synthetic, 3 seeds

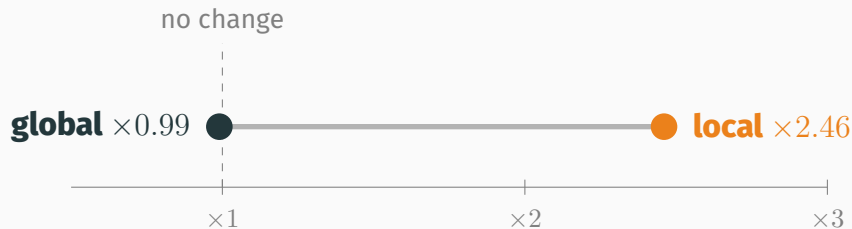
Real data: OPPORTUNITY



local wins **26 of 27** Eva, mean 0.211 vs 0.733

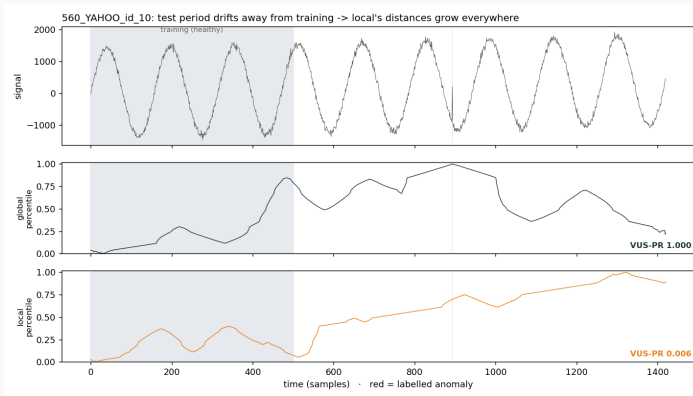
Healthy but unseen: a new amplitude

synthetic C2: healthy test data at an amplitude never seen in training



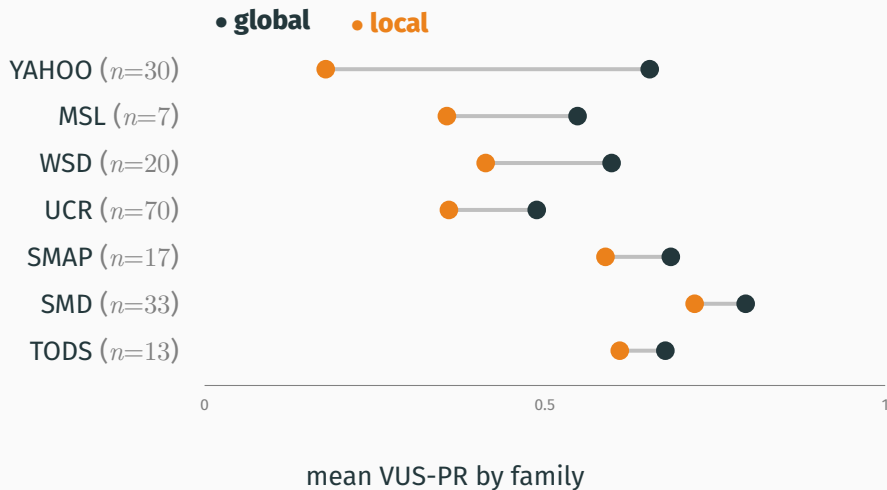
score rise on unseen healthy windows (median ratio)

Real data: drift after a short prefix

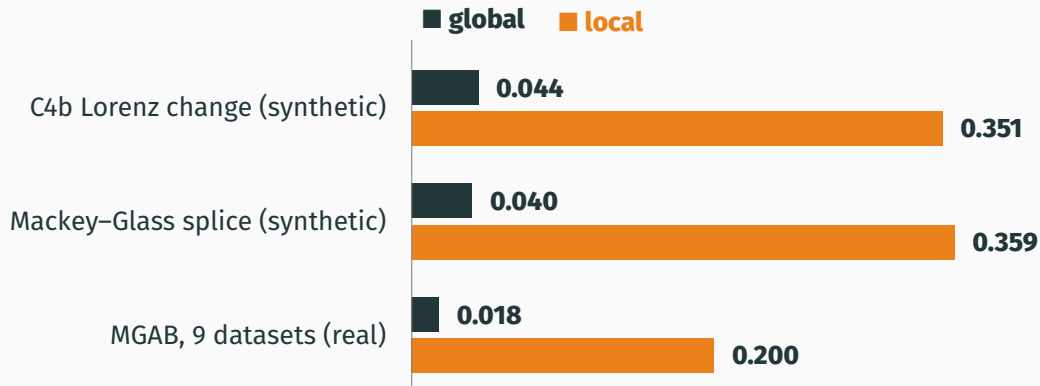


global **1.000** local **0.006** VUS-PR, 560_YAHOO_id_10

Where global wins on Eva



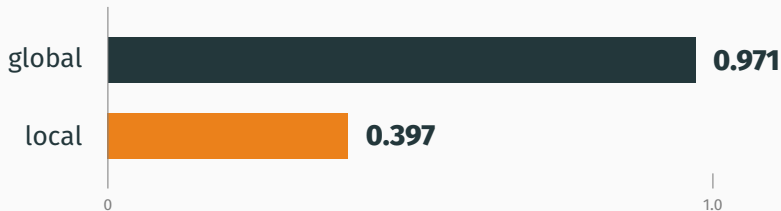
In span, off training set: a curved attractor



VUS-PR; axis 0 to 0.4

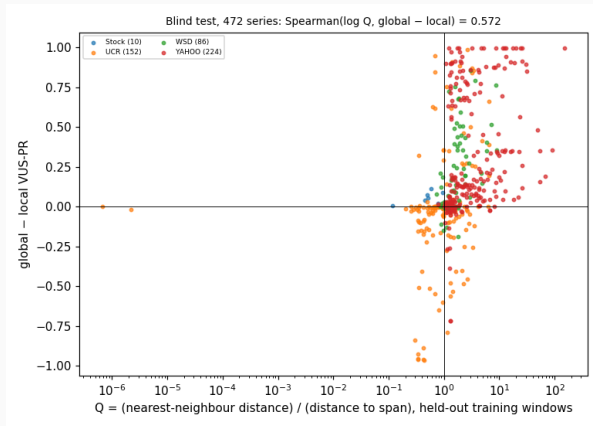
Counter-case: same attractor, global wins

C4: Lorenz + fast foreign oscillation (period 7)



“chaos favours local” **not supported**

Blind prediction from healthy data alone



$Q = r_{\text{NN}} / r_{\text{span}}$, no labels

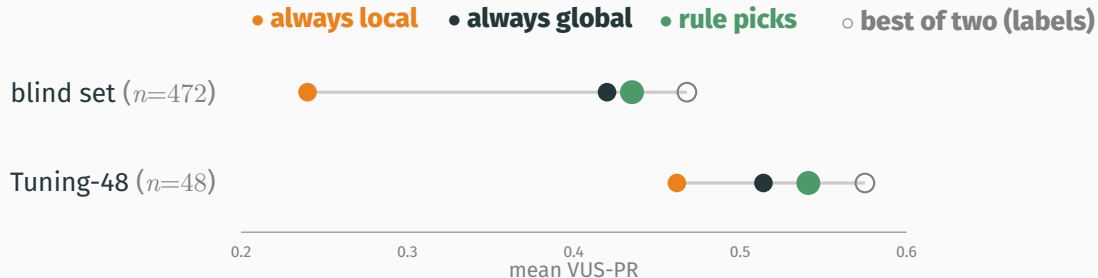
$$\rho = 0.572$$

accuracy **0.837**

always “global”: 0.784

family confound

Using the rule to choose the detector



rule vs always global: **+0.015** blind **+0.026** tuning

about a third of the possible gain

Where neither belief works

0.075 = **0.075**

global

local

142_MSL_id_3

4 / 27

OPPORTUNITY poor under local too

What is and is not established

checked

both identities

synthetic + real

beliefs predict the failures

partly

blind prediction ($\rho = 0.572$)

not established

mechanism on each real dataset

not new

Prop. 1 (Wiener / PPCA); Parzen windows

Reproducing

checks, figures	<code>utils/two_beliefs_report_figures.py</code>
noise sweep	<code>utils/scale_sweep_posterior.py</code>
synthetic cases	<code>utils/why_local_global_synthetic.py</code>
C2, C3 diagnostics	<code>utils/why_diagnostics_auditfix.py</code>
why_*.png	<code>utils/plot_why_local_global.py</code>